

# Project Synopsis

MudraTrack · GST Invoicing & Job-Card Management ERP

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|--------------|----------------------|---------------|------------|
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## 1. Project Overview

MudraTrack is an enterprise ERP and billing system built for print shops, digital packaging studios, and banner manufacturers. It connects job card production workflows directly with GST compliant invoicing, role-based access control, a prepaid wallet via Razorpay, automated WhatsApp and email notifications, and an AI assistant using Google Gemini for business reporting.

## 2. Technologies Used

- **Frontend:** React.js (v18), Redux Toolkit, Apollo Client, Material UI, Framer Motion
- **Backend:** NestJS, TypeScript, GraphQL, Prisma ORM, Node-cron, MongoDB

## 3. Work Done

### Frontend

- **Multi-Company Switching:** Built tenant switching with automatic Apollo Cache state synchronization.
- **Mobile Responsiveness:** Refactored dashboard pages, navigation and forms for all screen sizes.
- **Localization:** Added Marathi language support and fixed Hindi locale files. Localized entire project across all modules.
- **Wallet Recharge UI:** Developed WalletTab interface and integrated Razorpay Checkout SDK.
- **Bug Fixes:** Fixed state inconsistencies, form validation errors and screen rendering issues.

### Backend

- **Dynamic Pricing Engine:** Built volume-based slab discount calculator for automatic invoice totals.
- **Razorpay Integration:** Implemented payment order creation, SHA256 HMAC webhook verification and atomic wallet crediting.
- **WhatsApp Notifications:** Created WABA message dispatcher for order confirmations and receipt links.
- **Database Schema:** Designed MongoDB collections for wallet transactions and double-entry ledger using Prisma ORM.

## 4. Key Learnings

- Gained experience in full-stack ERP development using a professional monorepo setup with CI/CD.
- Learned secure payment gateway integration including cryptographic webhook signature validation.
- Implemented complete application localization in Marathi and Hindi.
- Worked with Google Gemini AI function-calling for natural language business intelligence queries.